

Task - JavaScript (ES6 + Loops + Higher Order Functions)

Part 1 - Choose

1) `map()` إيه اللي بيرجعه ؟

- ☐ أول عنصر يحقق شرط
 - ☒ جديدة بنفس الطول Array
 - ☐ Boolean
 - ☐ Number
-

2) مين فيهم بيرجع أول عنصر يحقق الشرط؟

- ☐ `filter()`
 - ☐ `map()`
 - ☒ `find()`
 - ☐ `forEach()`
-

3) `filter()` بيرجع...

- ☐ أول عنصر
 - ☒ جديدة بالعناصر اللي حققت الشرط Array
 - ☐ Number
 - ☐ String
-

4) `forEach()` بيرجع...

- ☐ جديدة Array
- ☐ أول عنصر
- ☒ undefined

- ☐ Boolean

5) for...of مع غالباً...

- ☐ Objects
- ☒ Arrays
- ☐ Functions
- ☐ Classes

Part 2 - True or False

1. map() الأصلية Array يغير ال. false
2. filter() فاضية Array ممكن يرجع true
3. find() ممكن يرجع undefined. true
4. for...in بتاع ال Index بيلف على ال true
5. break ينفع أعمل بيها false

Part 3 - Complete the following

Q1

خلي الكود يطبع:

```
2
4
6
8
const numbers = [1,2,3,4];

numbers.____Map____((num)=>{
  console.log(num * 2);
});
```

Q2

فيها الأرقام الأكبر من ٢٠ Array طلع

```
const nums = [10,25,5,30,15,40];

const result = nums.____filter____((num)=>{
  return num > 20;
});

console.log(result);
```

Q3

هات أول شخص عمره أكبر من ٢٥.

```
const users = [
  {name:"Ali", age:20},
  {name:"Sara", age:28},
  {name:"Omar", age:30}
];

const user = users.____find____((item)=>{
  return item.age > 25;
});

console.log(user);
```

Q4

Uppercase حوّل كل الأسماء لـ

```
const names = ["ali","mona","ahmed"];

const result = names.____map____((name)=>{
  return name.toUpperCase();
});

console.log(result);
```

Part 4 - To Do

دي Array عندك الـ

```
const fruits = ["Apple","Banana","Orange"];
```

1) for...of اطبع كل عنصر باستخدام

```
const fruits = ["Apple","Banana","Orange"];
for (const fruit of fruits){
```

```
console.log(fruits)
}
```

2) for...in باستخدام Index اطبع الـ

```
const fruits = ["Apple", "Banana", "Orange"];
for (const index in fruits){
    console.log(index);
}
```

3)

forEach اطبع بالشكل ده باستخدام

```
0 -> Apple
1 -> Banana
2 -> Orange
```

```
const fruits = ["Apple", "Banana", "Orange"];
fruits.forEach((fruit, index)=>{
    console.log(`${index} -> ${fruit}`);
});
```

Part 5 - To Do

Q1

Arrow Function حوّل الكود لـ

```
function sum(a,b){
    return a+b;
}
```

Solution:

```
let sum = (a, b) => {
    return a + b;
};
console.log(`sum is ${sum(5, 10)}`);
```

Q2

استخدم Destructuring

```
const user = {  
  name: "Mostafa",  
  age: 25  
};
```

في متغيرات age و name.

Solution:

```
const user = {  
  name: "Mostafa",  
  age: 25  
};  
const {name, age} = user;  
console.log(name);  
console.log(age);
```

Q3

استخدم Template Literal

بدل

```
console.log("Hello " + name);
```

Solution

```
console.log(`Hello + ${name}`);
```

Q4

استخدم Spread Operator

```
const arr1 = [1,2,3];  
const arr2 = [4,5,6];
```

Solution:

```
const arr1 = [1,2,3];  
const arr2 = [4,5,6];  
const newArr = [...arr1, ...arr2];  
console.log(newArr);
```

واحدة فيها الكل Array اعمل

Part 6 - Many Q

عندك البيانات دي

```
const students = [  
  {name:"Ali", degree:70},  
  {name:"Sara", degree:95},  
  {name:"Ahmed", degree:40},  
  {name:"Mona", degree:85},  
  {name:"Omar", degree:55}  
];
```

Required

1) فيها أسماء الطلبة بس Array اعمل

Solution:

```
const students = [  
  {name:"Ali", degree:70},  
  {name:"Sara", degree:95},  
  {name:"Ahmed", degree:40},  
  {name:"Mona", degree:85},  
  {name:"Omar", degree:55}  
];  
const names = students.map((student)=>{  
  return student.name;  
});  
console.log(names);
```

2) فيها الطلبة اللي درجاتهم أكبر من أو تساوي ٦٠ Array اعمل

```
const students = [  
  {name:"Ali", degree:70},  
  {name:"Sara", degree:95},  
  {name:"Ahmed", degree:40},  
  {name:"Mona", degree:85},  
  {name:"Omar", degree:55}  
];  
// const names = students.map((student)=>{  
//   return student.name;  
// });
```

```
// console.log(names);
const deg = students.filter((studnt) =>{
    return studnt.degree>=60;
});
console.log(deg);
```

3) هات أول طالب درجته أكبر من ٩٠

```
const students = [
    {name:"Ali", degree:70},
    {name:"Sara", degree:95},
    {name:"Ahmed", degree:40},
    {name:"Mona", degree:85},
    {name:"Omar", degree:55}
];
// const names = students.map((student)=>{
//     return student.name;
// });
// console.log(names);
const deg = students.find((studnt) =>{
    return studnt.degree>90;
});
console.log(deg);
```

4) اطبع أسماء كل الطلبة باستخدام forEach().

```
const students = [
    {name:"Ali", degree:70},
    {name:"Sara", degree:95},
    {name:"Ahmed", degree:40},
    {name:"Mona", degree:85},
    {name:"Omar", degree:55}
];
students.forEach((student)=>{
    console.log(student.name);
});
```

Bonus

عادية Loop بدون استخدام (for أو while)

MDN لو فاكرها أو دور عليها في الـ `reduce()` احسب مجموع الأرقام دي باستخدام

```
const numbers = [5,10,15,20];
```

Solution:

```
const numbers = [5,10,15,20];  
console.log( numbers.reduce( (sum, current) => {return sum + current},  
0 ) );
```